

Supplementary Material for *Credible on Paper? Evaluating the Effect of Green Banking Policy on Firms' Green Investment in Emerging Economies*

This file reports supplementary tables referenced in the main text: a small-sample supplementary check on a narrower outcome (Table S1), additional robustness checks and their multiple-testing correction (Tables S2–S3), the full per-economy SBFN and regulatory-quality coding underlying the primary sample (Table S4), Bayesian convergence diagnostics (Section S5), and the global sample's SBFN membership by region (Table S6). Section and table numbers here (S1, S2, ...) are internal to this supplementary file and distinct from the main text's own numbering, which this file's prose refers to descriptively (e.g., “the main text's baseline logit table”) rather than by number, since table numbers in the main text are subject to change on revision.

1 Small-sample supplementary check: waste-minimization adoption

Before the global-sample replication that is the paper's second major test, we report a smaller supplementary check using waste-minimization adoption—an outcome not captured in the standardized cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point—available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in the main text's baseline, hierarchical, and causal-forest results is a peculiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes: pooled across the four economies, having an overdraft facility is associated with significantly higher odds of waste-minimization adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green relationship replicates on a different continent, a different half-decade, and a narrower outcome definition than the one anchoring the main text's primary-sample results—a preview, on four economies, of the much larger replication in the global sample.

Table 1: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

2 Additional robustness checks

Table 2 subjects the primary-sample finding to four further checks, all retaining sector fixed effects, firm controls, and country-clustered standard errors (no country fixed effects, for the reason given in the main text). First, replacing the binary outcome with the continuous seven-item green-adoption index and estimating by OLS reproduces the same pattern to the decimal: credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistinguishable from zero ($b = 0.006$, $p = 0.739$).

Second, replacing bank credit access with two alternative finance measures produces one genuine point of nuance worth reporting rather than smoothing over. Using overdraft-facility access instead of a credit line/loan, the main overdraft effect remains positive and significant ($b = 0.274$, $p = 0.011$), the SBFN main effect remains negative and significant ($b = -0.905$, $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and significant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper that clears a conventional significance threshold, and it runs opposite to the direction H2 predicts. Using the reverse-coded perceived finance-obstacle measure in place of realized access, by contrast, both the main effect and its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interaction $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation noted in the main text. We read the overdraft result as suggestive rather than conclusive given it is one significant coefficient among more than a dozen interaction tests across the paper (see Table 3 below), but it is at least directionally consistent with a substantive story worth flagging for future work: if SBFN-guided lending is channelled preferentially through the term loans and credit lines that finance long-horizon capital expenditure, rather than through short-term overdraft facilities, we might expect exactly this pattern—a null-to-positive interaction on credit lines and a negative one on overdrafts—though our data cannot directly test whether SBFN-guided credit is disproportionately term-structured.

Third, splitting the sample into manufacturing ($N = 13,340$) and services ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main effect, and the null credit $\times$ SBFN interaction all replicate within each sector separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b = -0.040$, $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b = 0.195$, $p = 0.426$)—the paper’s central null is not an artifact of pooling two structurally different production technologies.

Because the overdraft interaction (the one coefficient in the paper that clears a conventional significance threshold) is one of many credit $\times$ institutional-moderator tests reported across the frequentist specifications in this paper, we additionally apply a Benjamini–Hochberg false-discovery-rate correction across the full family of 13 such coefficients reported

Table 2: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

under a classical (non-Bayesian, non-forest) estimator in the main text’s baseline, robustness, and global-sample regression tables (Table 3 below; full calculation in the replication package). The overdraft interaction’s raw $p = 0.031$ becomes $p = 0.403$ after correction, and no coefficient in the family survives at a false discovery rate of 0.10 or even 0.40. We read this as reinforcing, with a formal multiple-testing accounting rather than only a qualitative caveat, the suggestive-not-conclusive framing the main text’s Discussion already gives this result: an isolated nominally-significant interaction among more than a dozen tested is exactly what a pure false-discovery process would be expected to produce roughly one time in twenty, and this correction does not let us reject that account.

Table 3: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (main text’s baseline, robustness, and global-sample regression tables above). Entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the main text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table 2 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Main text, global sample, M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Main text, global sample, M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Main text, baseline M3 †	Credit $\times$ Reg. quality	0.310	0.799
Main text, global sample, M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table 2 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Main text, global sample, M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Main text, baseline M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table 2 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Main text, baseline M2	Credit $\times$ SBFN	0.773	0.892
Table 2 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table 2 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Main text, baseline M3 †	Credit $\times$ SBFN	0.892	0.892

3 SBFN policy status and regulatory quality, full sample

Table 4 reports, for all 47 economy-years in the paper’s primary and small-sample supplementary check, the SBFN membership status coded for the analysis, the join year where applicable, the SBFN Multi-Tiered Progress Framework stage in force nearest the survey

year, and the exact year of the Worldwide Governance Indicators observation used. Full per-country source citations (the specific SBFN Global Progress Report, country addendum, or accession announcement consulted for every determination) are available in the replication package. The analogous 162-economy table for the global sample is summarized by region in Table 5 below and provided in full in the replication package.

Table 4: SBFN policy status and regulatory quality, full 47-economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integration)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Preparation)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementation)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019
India 2022	1	2016	Developing (Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)	2023
Italy 2019	0	—	None	2019
Jordan 2019	1	2016	Commitment (Preparation)	2019
Kazakhstan 2019	0	—	None	2019
Kosovo 2019	0	—	None	2019
Kyrgyz Republic 2019	1	2018	Commitment (Preparation)	2019
Latvia 2019	0	—	None	2019
Lebanon 2019	0	—	None	2019
Lithuania 2019	0	—	None	2019
Malta 2019	0	—	None	2019
Moldova 2019	0	—	None	2019
Mongolia 2019	1	2012	Advancing (Implementation)	2019
Montenegro 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Morocco 2019	1	2014	Advancing (Implementa- tion)	2019
North Macedo- nia 2019	0	—	None	2019
Peru 2023	1	2013	Advancing (ESG Integra- tion)	2023
Philippines 2023	1	2013	Advancing (ESG Integra- tion)	2023
Poland 2019	0	—	None	2019
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Repub- lic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementa- tion)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

4 Bayesian model convergence diagnostics

Convergence diagnostics for every focal parameter of the hierarchical model reported in the main text are included directly in that table ($\hat{R}$ column); the full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member) to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional minimum threshold of 400 effective samples per parameter for stable posterior summaries at four chains. No divergent transitions were flagged during sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup draws each after 1,000 tuning iterations).

5 Global-sample SBFN status, regional summary

Of the 162 global-sample economy-years, 61 are coded as SBFN members as of their survey year and 101 as non-members. Table 5 reports the regional distribution of the 61 members, drawn from the SBFN data portal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste, Bangladesh, and Indonesia) carry the deeper per-country sourcing already reported in Table 4 above, having been determined during the primary sample’s original

research pass rather than from the portal roster alone.

Table 5: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61